

DP Computer Science: A 3.2.3 Data Types

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Outline the different data types used in relational databases
- Explain the importance of data type consistency
- Describe the potential effects of choosing the wrong data type

Key Vocabulary

Term	Definition
Data type	A category that defines what kind of data can be stored in a column of a database table
Relational database	A database that organises data into tables with rows and columns, linked by relationships
Consistency	Ensuring that all data in a column is of the same type, allowing reliable operations
INTEGER	A data type for storing whole numbers (positive and negative)
VARCHAR	A data type for storing variable-length text strings
DATE	A data type for storing calendar dates
BOOLEAN	A data type for storing true/false values
FLOAT/DECIMAL	Data types for storing numbers with decimal points

IB ATL Skills Covered

Skill	Description
Thinking	Critical thinking (analysing implications of data type choices); application (applying knowledge to database design)
Communication	Using appropriate terminology; explaining consequences of incorrect data type selection
Self-Management	Organisation (structuring data effectively using appropriate types)

2. What Are Data Types?

Definition

A data type defines what kind of data can be stored in a column of a database table. It sets rules for how data is stored, accessed, and manipulated.

Why Data Types Matter

text

WHY DATA TYPES MATTER

DATA TYPE DEFINES:

✓ What values are allowed (valid range)

✓ How much storage space is used

✓ What operations can be performed

✓ How data is sorted and compared

✓ How data is displayed

Without data types, the database would not know how to interpret, compare, or manipulate the stored data.

3. Common Data Types in Relational Databases

Numeric Data Types

Data Type	Description	Example	Use Case
INTEGER	Whole numbers (positive, negative, zero)	42 , -7 , 0	Student ID, Age, Counts
SMALLINT	Smaller whole numbers (saves space)	255 , -128	Short codes, flags
BIGINT	Very large whole numbers	9,223,372,036,854,775,807	Large IDs, timestamps
DECIMAL(p,s)	Exact decimal numbers (p = precision, s = scale)	19.99 , 3.14159	Currency, measurements
FLOAT	Approximate decimal numbers	3.14159 , 2.71828	Scientific calculations

Text/String Data Types

Data Type	Description	Example	Use Case
CHAR(n)	Fixed-length character string (always uses n characters)	'H' , 'JK'	Fixed codes, ZIP codes
VARCHAR(n)	Variable-length character string (up to n characters)	'Hello' , 'World'	Names, addresses, descriptions
TEXT	Variable-length with unlimited characters	'Long description...'	Longer text content

CHAR vs. VARCHAR:

Aspect	CHAR(n)	VARCHAR(n)
Length	Fixed (always uses n characters)	Variable (uses only the characters stored)
Storage	Wastes space for shorter strings	Efficient for variable-length data

Aspect	CHAR(n)	VARCHAR(n)
Performance	Faster (fixed length)	Slightly slower (variable length)
Best For	Fixed-length codes, IDs	Names, descriptions, addresses

Date and Time Data Types

Data Type	Description	Example	Use Case
DATE	Calendar date	'2026-07-17'	Birth dates, enrollment dates
TIME	Time of day	'14:30:00'	Class start times
TIMESTAMP	Date and time	'2026-07-17 14:30:00'	Log entries, transactions

Boolean Data Type

Data Type	Description	Example	Use Case
BOOLEAN	True/false values	TRUE , FALSE	Is enrolled? Is active?

Binary/Other Data Types

Data Type	Description	Use Case
BLOB	Binary Large Object (images, files)	Storing images, PDFs, documents
ENUM	Set of predefined values	Gender, status codes

4. Data Type Consistency

What is Consistency?

Data type consistency means that all data in a column of a database table is stored using the same data type. This ensures reliable operations, accurate comparisons, and efficient storage.

Why Consistency Matters

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WHY CONSISTENCY MATTERS

Without Consistency:

✗ Sorting becomes unpredictable

("10" would come before "2" if stored as text)

✗ Mathematical operations fail

(You cannot add "10" and "20" if stored as text)

✗ Comparisons give unexpected results

(Which is bigger: "100" or "20"?)

❌ Storage is wasted (Using VARCHAR(255) for a 2-digit number)
❌ Queries produce incorrect results (Searching for "10" may not find "10")

Scenario Example: Sorting Inconsistency

Scenario: A database stores student ages as VARCHAR instead of INTEGER.

Student	Age (Stored as VARCHAR)	Expected Sort Order	Actual Sort Order
Alice	"8"	Alice (8)	Alice (8)
Bob	"10"	Charlie (10)	David (10)
Charlie	"15"	Bob (15)	Bob (15)
David	"10"	David (10)	Charlie (15)

Problem: When sorted as text, "10" comes before "8" because the first character '1' is less than '8'. This is alphabetically correct but numerically wrong!

5. Effects of Choosing the Wrong Data Type

What Can Go Wrong?

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EFFECTS OF WRONG DATA TYPE CHOICE
1. DATA LOSS - Decimal values are truncated - Text beyond length limit is cut off - Dates outside range are invalid
2. DATA CORRUPTION - Text stored in numeric fields - Invalid date formats - Malformed data
3. STORAGE WASTE - Using VARCHAR(255) for short strings - Using BIGINT when INTEGER would suffice - Inefficient storage allocation
4. APPLICATION ERRORS - Calculations produce wrong results - Sorting and filtering are incorrect - Queries fail

Bad Practice Scenarios

Scenario	Wrong Choice	Consequence	Correct Choice
Storing ages	VARCHAR	Sorting errors ("10" before "8")	INTEGER
Storing prices	VARCHAR	Cannot perform arithmetic	DECIMAL(10,2)
Storing phone numbers	INTEGER	Leading zeros lost	VARCHAR
Storing dates	VARCHAR	Invalid date comparisons	DATE
Storing student names	CHAR(5)	Names cut off	VARCHAR(50)

6. Data Type Selection Guide

Choosing the Right Type

Data	Recommended Type	Reason
Student ID	INTEGER	Whole numbers, sequential
Name	VARCHAR(50)	Variable length, text
Birth Date	DATE	Date operations needed
GPA	DECIMAL(3,2)	Exact decimal, consistent precision
Is Enrolled?	BOOLEAN	True/false values
Address	VARCHAR(255)	Variable length text
Phone Number	VARCHAR(15)	Leading zeros, special characters
Description	TEXT	Longer text content
Postal Code	VARCHAR(10)	Leading zeros, varying lengths

Guidelines for Selection

Rule	Description
Match the Data	Choose a type that accurately represents the data being stored
Be Specific	Use the most specific type available (DATE, not VARCHAR)
Avoid Over-sizing	Use the smallest type that fits the data to save space
Consider Operations	What operations will be performed? (Sorting, arithmetic, comparison)
Handle Special Cases	Leading zeros, decimal precision, time zones
Plan for Growth	Allow enough space for future values

7. Lesson Sequence

Lesson Plan

Phase	Time	Activity
Introduction	5 min	"What types of information do we store about a student?"; introduce data types
Data Types	15 min	Core data types (INTEGER, VARCHAR, DATE, BOOLEAN); fast-paced Q&A
Consistency	15 min	Think-Pair-Share: "What if" scenarios (sorting text as numbers)
Wrong Types	15 min	Group brainstorm: bad-practice scenarios and consequences
Wrap-up	10 min	Rapid-fire Q&A; exit tickets

Activity: Think-Pair-Share — Consistency Scenarios

Scenario 1: Sorting Ages

- **Problem:** Ages stored as VARCHAR
- **Question:** What happens when you sort the ages?
- **Answer:** "10" comes before "8" (alphabetical, not numerical)

Scenario 2: Adding Prices

- **Problem:** Prices stored as VARCHAR
- **Question:** What happens when you try to calculate total sales?
- **Answer:** Arithmetic operations fail; numbers treated as text

Scenario 3: Searching Names

- **Problem:** Names stored as CHAR(10)
- **Question:** What happens to names longer than 10 characters?
- **Answer:** Names are truncated; data loss occurs

Activity: Group Brainstorm — Wrong Type Consequences

Group 1: Email Column

- **Wrong Type:** CHAR(10) → **Correct Type:** VARCHAR(100)
- **Consequence:** Email addresses are cut off; data loss

Group 2: Student Grade Column

- **Wrong Type:** VARCHAR → **Correct Type:** DECIMAL(4,2)
- **Consequence:** Cannot calculate averages; sorting errors

Group 3: Phone Number Column

- **Wrong Type:** INTEGER → **Correct Type:** VARCHAR(15)
- **Consequence:** Leading zeros lost; cannot store special characters

Group 4: Date of Birth Column

- **Wrong Type:** VARCHAR → **Correct Type:** DATE
- **Consequence:** Cannot calculate age; inconsistent date formats

Group 5: Salary Column

- **Wrong Type:** VARCHAR → **Correct Type:** DECIMAL(10,2)
 - **Consequence:** Cannot perform arithmetic; sorting errors
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8. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
Think-Pair-Share	Assess understanding of consistency implications
Group Brainstorm	Evaluate ability to identify consequences of wrong types
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. What is the difference between CHAR and VARCHAR?
 2. Which data type would you use for storing student ages? Why?
 3. What happens if you store a price in a VARCHAR column instead of DECIMAL?
 4. Why is data type consistency important?
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9. Differentiation

Level	Strategy
Support	Provide data type reference sheet; use visual aids and real-world examples
Challenge	Encourage research on more complex data types (BLOB, ENUM, JSON) or performance implications

10. Resources

- **Presentation** (Slide Deck)
 - **eBook** (A3.2.3 with key questions and quiz)
 - **Worksheet:** Data type scenarios and consequences
 - **Quizlet** (A3.2.3 vocabulary flashcards)
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11. Summary: Key Takeaways

Concept	Key Point
INTEGER	Whole numbers (age, ID, counts)
VARCHAR	Variable-length text (names, addresses)
CHAR	Fixed-length text (codes, ZIP codes)
DATE	Calendar dates (birth dates, enrollments)
DECIMAL	Exact decimal numbers (prices, measurements)

Concept	Key Point
BOOLEAN	True/false values (is enrolled?)
Consistency	Ensures reliable sorting, operations, and storage
Wrong Type	Can cause data loss, corruption, storage waste, and application errors

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KEY REMINDERS	
✔	Choosing the right data type is essential
✔	Consistency ensures reliable operations
✔	Wrong type → data loss, corruption, errors
✔	Use the most specific type available
✔	Avoid over-sizing (wastes storage)

"Choosing the right data type is essential for database accuracy, efficiency, and reliability—the wrong choice can lead to data loss, incorrect results, and wasted storage."